TECHNICAL REFERENCE GUIDE

interactive_toy_objects_MTObjects

Methodology, controls, parameters, and interpretation

Purpose: Interactively assess how MTObjects masks an unaltered S4G image and how reliably it recovers one or more synthetic foreground objects under controlled morphology, brightness, and placement.

Companion documentation for interactive_toy_objects_MTObjects.py

1. Scope and scientific purpose

The application is an interactive validation environment for the MTObjects foreground-masking methodology. It combines a real S4G 3.6 µm galaxy image with optional synthetic objects and displays the resulting segmentation, cleaned image, and recovery classification. It can also run with no toy object, providing a baseline measure of masking aggressiveness on the unaltered image.

2. Processing methodology

1. Select a machine and galaxy. The FITS image and catalogue geometry are loaded.
2. Calibrate the MTObjects background variance from the robust noise of a large-scale-subtracted residual image.
3. Optionally define and place one or more toy objects in deprojected, bar-aligned coordinates.
4. Add all committed and pending toy models to a copy of the galaxy image.
5. Run MTObjects: background subtraction, Gaussian smoothing, max-tree construction, statistical node filtering, segment relabelling, and post-filtering.
6. Dilate retained segments, replace masked pixels in the recovered image, and compare connected mask regions with the known toy truth masks.

Baseline mode: Press Calculate with no placed toys to run MTObjects on the original image. This is the recommended first step when judging false-positive masking and tuning aggressiveness.

3. Toy object specification

Control	Meaning	Effect
Toy type	Gaussian star, star cluster, or compact galaxy.	Selects the synthetic morphology.
x / y deprojected arcsec	Position in the deprojected, bar-aligned frame.	Mouse clicks update these coordinates.
Peak residual sigma	Peak amplitude in units of robust image sigma.	Higher values make the toy brighter; initial value is 30.
FWHM pixels	Characteristic width of the toy.	Higher values make it broader.

Control	Meaning	Effect
Axis ratio	Minor-to-major axis ratio.	Used by compact galaxies; 1 is circular.
Object PA deg	Toy position angle in the observed image.	Controls compact-galaxy orientation.
Truth dilation pixels	Expansion of the truth footprint.	Allows tolerance around the analytic toy boundary.

4. Placing multiple toys

- Click the top-left image to create a pending toy at the cursor position; the red cross marks it.
- Add toy commits the current type, position, brightness, shape, and truth dilation.
- Committed toys appear as numbered blue circles. Change the controls and repeat to add different toys.
- Remove last deletes the most recently committed toy. Clear removes all committed and pending toys.
- Calculate processes all committed toys plus any pending toy at the red cross.

5. MTObjects parameter reference

An arrow shows the direction that generally produces more masking. “Variable” means the response is image-dependent. The Reset button restores the initial values and recalibrates background variance for the selected galaxy.

Parameter	Initial value	More mask	Description
detect_on	original	Variable	Runs on the original image or a large-scale-subtracted residual.
alpha	1e-6	Fixed	Significance level. The bundled original significance test

Parameter	Initial value	More mask	Description
			requires 1e-6.
move_factor	0.7839	↓	Moves object markers through the max-tree. Lower values retain broader/fainter outskirts.
min_distance	0.455	↓	Minimum normalized brightness separation required between objects.
gaussian_fwhm	3.92 px	Variable	Pre-detection smoothing scale. May improve diffuse detection but merge or suppress compact sources.
soft_bias	0	Variable	Bias term used during gain estimation and noise modelling.
gain	−1	Variable	Electrons per ADU; −1 requests estimation.
bg_mean	NaN	Variable	Background mean; NaN requests estimation.
bg_variance	per galaxy	↓	Noise variance used by the significance test; calibrated from robust residual noise.
minarea	79 px	↓	Rejects retained

Parameter	Initial value	More mask	Description
			segments smaller than this area.
dilation_radius	2 px	↑	Expands every retained mask region.
max_area	2485 px	↑	Rejects segments larger than this area.
max_elongation	10.21	↑	Rejects segments more elongated than this ratio.
exclude_center_pixels	8 px	↓	Rejects segments whose centroid lies inside the central exclusion radius.

6. Background calibration

MTObjects tests whether max-tree nodes contain statistically significant flux relative to the background. Its variance must therefore be expressed in the same numerical scale as the FITS image. A variance copied from differently scaled optimisation data can make the assumed noise enormously too large and suppress every raw detection.

Implemented safeguard: For each selected galaxy, the application subtracts a 15-pixel Gaussian large-scale model, measures robust residual sigma using 1.4826 × MAD, and sets bg_variance to sigma^2. The displayed background RMS is the square root of the variance used by MTObjects.

For ESO120-012, an earlier fixed variance of 6583 implied RMS ≈ 81 and produced zero raw labels. Per-image residual calibration gave variance $\approx 2.98\times10^{-4}$ (RMS ≈ 0.0173), restoring detections. Values vary by galaxy and should not be copied blindly between image units or processing stages.

7. Output panels

Panel	Contents	Interpretation
Toy position	Original galaxy, committed toy positions, and pending red cross.	Placement view; no synthetic flux is displayed here.
Injected toy	Galaxy plus the actual summed toy models.	Confirms morphology, brightness, and position.
MTObjects mask	Binary post-filtered and dilated mask.	Black regions are removed by the configured pipeline.
Recovered image	Cleaned image with classification outlines.	Shows the practical masking result and errors.

8. Outline classification

- Green: the portion of a connected detected region that correctly covers toy truth.
- Red around a green outline: the full outer boundary of that connected detection when it extends beyond the toy truth footprint.
- Red without green: a masked connected region that overlaps no toy and is therefore an incorrect removal for the toy-recovery experiment.

Connected-region classification avoids the misleading nested red contours that occur when excess pixels are contoured as a hollow annulus.

9. Status metrics

Metric	Definition
Recovered truth pixels	Number and percentage of truth pixels covered by the mask (pixel recall).
Kept/raw segments	Post-filter retained segment count divided by the raw MTObjects segment count.

Metric	Definition
Masked fraction	Fraction of all image pixels in the final dilated mask.
Background RMS	Noise scale used in MTObjects statistical significance calculations.

10. Recommended tuning sequence

1. Run the unaltered baseline and inspect false-positive red outlines and masked fraction.
2. Add a clearly visible toy (30σ is the initial value) in a representative location.
3. Confirm background RMS is plausible for the image scale before changing detection parameters.
4. Reduce minarea if compact toys are detected by MTObjects but rejected by post-filtering.
5. Try move_factor = 0.5, then 0.3, to retain fainter outskirts; monitor increased over-masking.
6. Adjust dilation only after the underlying segmentation is satisfactory.
7. Test several toy types, positions, and brightnesses, then repeat across galaxies.

11. Caveats

- A green outline indicates overlap, not perfect segmentation. The red outer outline quantifies spatial overreach visually.
- A toy may merge with a pre-existing source; connected-component classification then treats the combined region as one detection.
- Parameter effects are not always monotonic because max-tree topology changes with smoothing, background estimates, and source crowding.
- Optimised means are dataset-specific. Background statistics are especially sensitive to image units and preprocessing.

12. Implementation and source basis

The launcher interactive_toy_objects_MTObjects.py delegates to Foreground Masking/Shared/toy_object_interactive_core.py and uses the bundled mtobjects/mtolib implementation. The original MTObjects README recommends alpha = 1e-6, move_factor = 0.5 as its baseline, and move_factor in the range 0.0–0.3 when including faint outskirts is important. The current interactive initial values retain the project's shared MTObjects defaults, except that background variance is calibrated per galaxy for numerical consistency.

Scientific background: Haigh, C. et al. (2021), "Optimising and comparing source-extraction tools using objective segmentation quality criteria," Astronomy & Astrophysics, 645, A107. DOI: 10.1051/0004-6361/201936561.